at the age of 18, he entered the Augustinian Monastery at Brunn as a priest. From there he went to the University of Vienna for training in physics, mathematics and natural science. Mendel returned to the monastery in 1854 and continued to work as a priest and teach in high school. In his leisure time he started his famous experiments on the garden pea plant. He conducted his experiments in the monastery for about seven years from 1856 to 1863. He had worked on nearly 10000 pea plants of 34 different varieties. Mendel noted that they differ from one another in many ways.

Thus Mendel had chosen 7 pairs of contrasting characters for his study as shown in the table.

Table 18.1 Constrasting characters of pea plant used by Mendel

Characteristic studied	Dominant character	Recessive character
Stem length	Tall	Dwarf
Flower Position	Axillary	Terminal
Flower colour	Blue	White
Pod shape	Inflated	Constricted
Pod colour	Green	Yellow
Seed shape	Round	Wrinkled
Seed colour	Yellow	Green

Reasons for Mendel's success

He chose the pea plant as it was advantageous for experimental work in many aspects

1. It is naturally self-pollinating and so is very easy to raise pure breeding individuals.
2. It has a short life span as it is an annual and so it was possible to follow several generations.
3. It is easy to cross-pollinate.
4. It has deeply defined contrasting characters.
5. The flowers are bisexual.

18.2 Monohybrid Cross – Inheritance of One Gene

Crosses involving **inheritance of only one pair of contrasting characters** are called monohybrid crosses. For example it is a cross between two forms of a single trait like a cross between tall and dwarf plant.

Mendel's Explanation of Monohybrid Cross

Parental generation: Pure breeding tall plant and a pure breeding dwarf plant.

F_1 generation: Plants raised from the seeds of pure breeding parental cross in F_1 generation were tall and monohybrids.

F_2 generation: Selfing of the F_1 monohybrids resulted in tall and dwarf plants respectively in the ratio of 3:1. The actual number of tall and dwarf plants obtained by Mendel was 787 tall and 277 dwarf. External expression of a particular trait is known as phenotype. So the phenotypic ratio is 3:1.

In the F_2 generation 3 different types were obtained:

Tall Homozygous – TT (Pure) – 1
Tall Heterozygous – Tt – 2
Dwarf Homozygous – tt – 1

So the **genotypic ratio 1:2:1**. A genotype is the genetic constitution of an organism

Mendel's Interpretation on Monohybrid cross

Based on these observations it was confirmed by Mendel that 'factors' are passed on from one generation to another, **now refered to as genes**. Tallness and Dwarfness are determined by a pair of contrasting factors , tall plant possess a pair of factors (represented by T- taking the first letter of the dominant character) and a plant is dwarf because it possess factors for dwarfness (represented as t- recessive character). These factors occur

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